

Title

Protocol for evaluating community acceptance and efficacy of novel spatial repellents for malaria prevention

Abstract

Background: Despite distribution of insecticide-treated nets, malaria transmission persists in sub-Saharan Africa due to insecticide resistance and outdoor biting. Spatial repellents offer a complementary approach. This study protocol outlines a cluster-randomized trial evaluating a novel transfluthrin-based spatial repellent product.

Methods: The trial will be conducted in rural Kenya across matched clusters of households with children aged 6 months to 10 years. Clusters are randomly assigned to receive the novel spatial repellent product or standard prevention guidance. The intervention involves monthly product installation and replacement. Primary efficacy outcome: malaria infection incidence in children aged 6-10 years over 12 months of follow-up, assessed via blood smear and rapid diagnostic testing at baseline, 6 months, and 12 months. Secondary outcomes: community perceptions of product utility, ease of installation, perceived mosquito reduction, and comparison with existing prevention methods. Process evaluation: qualitative interviews at 1 week and 2 months post-installation to assess early user experience and identify implementation barriers. Blinding: field teams collecting clinical data will be masked to cluster assignment. Study design: double-blind cluster-randomized controlled trial.